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| Reg. No. |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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**B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025**

Third Semester

**21CSS202T - FUNDAMENTALS OF DATA SCIENCE**

*(For the candidates admitted from the academic year 2021-2022 to 2024-2025)*

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

**PART - A (20 x 1 = 20 Marks)**

Answer ALL Questions

Marks BL CO PO

- |                                                                                                                       |   |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 1. Which of the following is NOT a typical benefit of using data science in business?                                 | 1 | 1 | 1 | 1 |
| A) Improved decision-making based on data insights                                                                    |   |   |   |   |
| B) Increased data storage costs                                                                                       |   |   |   |   |
| C) Enhanced customer experience through personalized recommendations                                                  |   |   |   |   |
| D) Optimization of operational efficiency                                                                             |   |   |   |   |
| 2. Which of the following facets is considered a major type of data?                                                  | 1 | 1 | 1 | 1 |
| A) Structured                                                                                                         |   |   |   |   |
| B) Abstract                                                                                                           |   |   |   |   |
| C) Conceptual                                                                                                         |   |   |   |   |
| D) Synthetic                                                                                                          |   |   |   |   |
| 3. Which of the following Python libraries is primarily used for numerical operations and array manipulation?         | 1 | 1 | 1 | 1 |
| A) Pandas                                                                                                             |   |   |   |   |
| B) Matplotlib                                                                                                         |   |   |   |   |
| C) Numpy                                                                                                              |   |   |   |   |
| D) Scikit-learn                                                                                                       |   |   |   |   |
| 4. What function would you use to concatenate two numpy arrays along a specified axis?                                | 1 | 2 | 1 | 2 |
| A) numpy.split()                                                                                                      |   |   |   |   |
| B) numpy.join()                                                                                                       |   |   |   |   |
| C) numpy.concatenate()                                                                                                |   |   |   |   |
| D) numpy.append()                                                                                                     |   |   |   |   |
| 5. Which of the following commands is used to start the Python Debugger (PDB) in an interactive session?              | 1 | 2 | 2 | 3 |
| A) python -m pdb                                                                                                      |   |   |   |   |
| B) import pdb; pdb.set_trace()                                                                                        |   |   |   |   |
| C) pdb.start()                                                                                                        |   |   |   |   |
| D) debugger.start()                                                                                                   |   |   |   |   |
| 6. What Python module is commonly used for developing unit tests?                                                     | 1 | 2 | 2 | 1 |
| A) unittest                                                                                                           |   |   |   |   |
| B) debugger                                                                                                           |   |   |   |   |
| C) testlib                                                                                                            |   |   |   |   |
| D) pdb                                                                                                                |   |   |   |   |
| 7. Which method reads a JSON file and converts it into a Python dictionary?                                           | 1 | 2 | 2 | 2 |
| A) json.parse()                                                                                                       |   |   |   |   |
| B) json.load()                                                                                                        |   |   |   |   |
| C) json.read()                                                                                                        |   |   |   |   |
| D) json.decode()                                                                                                      |   |   |   |   |
| 8. Which function in the re module is used to replace occurrences of a pattern in a string with a replacement string? | 1 | 2 | 2 | 2 |
| A) re.findall()                                                                                                       |   |   |   |   |
| B) re.search()                                                                                                        |   |   |   |   |
| C) re.split()                                                                                                         |   |   |   |   |
| D) re.sub()                                                                                                           |   |   |   |   |

9. What is the primary purpose of Django's URLConf? 1 1 3 1  
 A) To define the layout and structure of the web pages  
 B) To handle HTTP requests and responses  
 C) To map URL patterns to views  
 D) To manage database connections
10. When using Django's template system, which tag is used to render a variable within a template? 1 2 3 1  
 A) {% if %}  
 B) {{ variable }}  
 C) {% for %}  
 D) {% block %}
11. Which method should be used to handle form submission and validate form data in Django? 1 2 3 2  
 A) form.is\_valid()  
 B) form.validate()  
 C) form.save()  
 D) form.check()
12. What is the role of Django's Model class in the MVC model? 1 1 3 1  
 A) It manages the URL routing  
 B) It handles HTTP requests and responses  
 C) It defines the data structure and provides methods to interact with the database  
 D) It generates HTML templates
13. What is one of the primary issues faced when handling large datasets in data analysis? 1 2 4 2  
 A) Difficulty in plotting data  
 B) Limited number of available libraries  
 C) High memory consumption and performance bottlenecks  
 D) Limited options for statistical analysis
14. Which Pandas function is used to create a DataFrame from a dictionary of lists? 1 2 4 1  
 A) pd.read\_csv()  
 B) pd.DataFrame()  
 C) pd.Series()  
 D) pd.concat()
15. How can you perform arithmetic operations on a DataFrame in Pandas? 1 2 4 1  
 A) Use the apply() function  
 B) Use direct arithmetic operators (+, -, \*, /) on the DataFrame  
 C) Use the map() function  
 D) Use the groupby() function
16. In Pandas, how do you handle missing data in a DataFrame? 1 2 4 2  
 A) Use pd.fillna() to replace missing values  
 B) Use pd.dropna() to remove rows with missing values  
 C) Use both pd.fillna() and pd.dropna()  
 D) Use pd.isnull() to identify missing data
17. What is one of the primary advantages of using Seaborn over Matplotlib for statistical plots? 1 1 5 1  
 A) Seaborn has more extensive data manipulation capabilities  
 B) Seaborn provides built-in support for complex statistical plots and integrates well with Pandas  
 C) Seaborn offers faster execution times for plotting  
 D) Seaborn requires less coding to create basic plots
18. Which Matplotlib function is typically used to create a scatter plot? 1 2 5 1  
 A) plt.plot()  
 B) plt.hist()  
 C) plt.scatter()  
 D) plt.bar()
19. Which of the following graphs would be most appropriate for visualizing the distribution of a single numeric variable? 1 2 5 1  
 A) Box plot  
 B) Heatmap  
 C) Pair plot  
 D) Bar plot

20. To analyze the relationship between two categorical variables, which type of plot is most suitable? 1 2 5 1
- A) Scatter plot B) Histogram  
C) Heatmap D) Line plot

**PART - B (5 x 8 = 40 Marks)**  
Answer ALL the Questions

Marks BL CO PO

- 21 a. Explain the process of creating and manipulating numpy arrays. Include examples of how to create arrays, perform basic operations (such as join, split, search, and sort), and manipulate array shapes. Provide a detailed explanation and code snippets where necessary. 8 3 1 3

(OR)

- b. Discuss the characteristics and applications of the Normal distribution, and provide Python code example. 8 3 1 3
- 22 a. Explain how to debug Python scripts using PDB and an Integrated Development Environment (IDE). Additionally, discuss the process of developing unit tests in Python. Include examples of common debugging commands in PDB and basic unit testing code. 8 3 2 3

(OR)

- b. Explain how to read, write, and parse JSON files in Python, including conversions between JSON and Python dictionaries. Provide code examples for each operation. 8 3 2 3
- 23 a. Explain the MVC (Model-View-Controller) model in the context of Django. How does Django implement this pattern? 8 4 3 4

(OR)

- b. Describe how you can handle form data in Django. Include details on creating forms, processing form submissions, and validating form data. 8 4 3 3
- 24 a. Explain the strategies for handling large datasets in Pandas, and discuss techniques for optimizing performance and memory usage. 8 4 4 4

(OR)

- b. Describe the usage of Pandas for time series data analysis, including handling date functionality, time deltas, and vectorization concepts. 8 4 4 3
- 25 a. Discuss the advantages of using Matplotlib and Seaborn for data visualization. Include their use cases and how they complement each other. 8 4 5 3

(OR)

- b. Explain how to choose appropriate graphical techniques for data visualization. Discuss the types of graphs suitable for univariate, bivariate, and multivariate data, and provide examples. 8 4 5 3

**PART - C (1 x 15 = 15 Marks)**  
Answer ANY ONE Questions

Marks BL CO PO

- 26 Suppose a retail company want to predict which products will be the most popular during the holiday season. Explain the data science process for the above mentioned scenario. 15 3 1 3
- 27 Explain the need for the Regular Expression. Discuss the Python code for performing each of the following basic functions: findall(), search(), split(), sub() 15 3 2 3

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